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# Proposal Revision Summary

Major Changes from Submitted Midway Proposal to Final Updated Version

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# Agenda

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1. Why the proposal had to be revised
2. Motivation and research-question changes
3. Methodology and chosen-approach changes
4. Experiment and statistics changes
5. Related-work and reference corrections
6. What stayed constant across versions

# Why We Revised the Proposal

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## Problems in the submitted version

- Some methods and references were weakly grounded or mismatched to the actual study plan.
- The original design mixed domains in ways that made interpretation harder.
- Several sections sounded over-polished and AI-like.
- A few methodological choices were not realistic for the intended scope and implementation path.

## Revision goals

- Keep the same core research problem.
- Remove hallucination-prone or unsupported claims.
- Make the design easier to justify and implement.
- Tighten the writing into a clearer bachelor-level proposal.

**Bottom line:** the final version is the same project, but reframed into a more defensible and implementable form.

# Motivation and Research Questions

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## Original version

- Stronger rhetorical emphasis on security, science, and practice.
- More expansive wording about mixed real/synthetic traffic.
- RQs were tied to a less stable methodology.

## Updated version

- Motivation now starts from steganography and steganalysis as one problem.
- The core question is unchanged.
- RQs are labeled by role: primary, exploratory, and verification.

Main change: tighter framing without changing the actual problem.

# Chosen Approaches: Major Method Changes

| Area                | Submitted version                               | Updated version                                                    |
|---------------------|-------------------------------------------------|--------------------------------------------------------------------|
| Carrier format      | RGB PNG everywhere                              | Grayscale throughout; PNG for spatial, JPEG Q=95 for DCT           |
| Quality control     | BRISQUE gate + artifact rejection               | Saturation + semantic screening; BRISQUE discussed, not used       |
| Spatial embedding   | PRNG-keyed pixel selection, $k = 1/2$ framing   | Sequential row-major LSB, with payload levels defined by fill rate |
| Frequency embedding | DCT-QIM on PNG carriers                         | DCT-LSB on actual JPEG carriers via quantized coefficients         |
| Domain comparison   | Method contrast entangled with carrier handling | Full pipeline contrast: LSB+PNG vs DCT-LSB+JPEG                    |
| Detector scope      | SRM+EC primary baseline                         | Training-free primary detectors; SRNet optional                    |

## Why this matters

These changes removed the most fragile parts of the original design and made the proposal match a realistic implementation path.

# Detectors and Statistical Evaluation

## Submitted version

- Detector set centered on SRM+EC plus branch-specific checks.
- Statistics relied on Wilcoxon, ANOVA, and Cohen's  $d$ .
- Trained and training-free detectors were mixed in the main analysis.

## Updated version

- Primary analysis now uses RS,  $\chi^2$ , Sample Pairs, and calibration  $\chi^2$ .
- ROC-AUC remains the metric, but comparisons now use DeLong for correlated AUCs.
- Multiple testing is handled explicitly with Bonferroni–Holm.
- SRNet remains possible, but only as a separate extension.

## Key methodological correction

The updated detector/statistics stack fits the actual study structure much better than the original mixed setup.

# Related Work and Reference Corrections

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## What was problematic before

- The original proposal used a broader but less stable reference set.
- It treated De et al. as the closest prior work.
- QIM-based framing and some detector references were tied to the dropped methodology.

## What the final version does

- Related work is now built around the papers that directly support the final design.
- Gao is used to distinguish coverless/generative steganography from our setting.
- Méreur is used as the closest prior comparison on AI-generated carriers.
- Mallet, Giboulot, and Corvi anchor the cover-source mismatch and synthetic-image forensics argument.

Main change: the literature review became narrower, but more accurate and better aligned with the actual proposal.

# What Stayed Constant

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## Core study idea

- The project still asks whether carrier origin changes detectability.
- The main comparison is still real vs. ML, with ML-A vs. ML-B as a second contrast.
- The overall scale remains 1,500 images across 3 sources.
- Payload, encryption, and two embedding branches remain part of the design.

## Practical takeaway for the team

- We did **not** abandon the project direction.
- We removed risky claims and unrealistic components.
- The final document is closer to what we can build and defend.



# Net Effect of the Revision

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## Improved credibility

- fewer hallucination risks
- cleaner literature positioning
- more defensible claims

## Improved methodology

- branch-consistent carriers
- more realistic detectors
- statistics matched to AUC comparisons

## Improved communication

- less AI-like wording
- clearer flow
- tighter explanation of scope and limitations

## Team summary

The final proposal is best understood as a correction and sharpening of the original submission, not a change of direction.